

Task-1

No.	BIOS/POST Error Message	Usually Indicates
1	CMOS Checksum Error	CMOS/BIOS settings may be corrupted or the CMOS battery may be weak/dead .
2	Keyboard Error / Keyboard Not Detected	Keyboard is disconnected, faulty, or the USB/PS/2 port has a problem.
3	Memory Error / RAM Error	RAM is loose, incompatible, faulty, or improperly installed.
4	No Boot Device Found	The HDD/SSD is not detected , the boot drive is faulty, or the BIOS boot order is incorrect.
5	CPU Fan Error	The CPU fan is disconnected, not spinning, blocked by dust, or failing ; BIOS may also have an incorrect fan-monitoring setting.

Task-2

Answer

For a traditional **AMI (American Megatrends) BIOS**, **3 short repeating beeps** typically indicate a **base memory (first 64 KB of RAM) failure**. This means the POST cannot properly read/write to system memory, so the PC may stop before initializing the display.

Two troubleshooting steps:

1. **Reseat the RAM:** Power off and unplug the PC, remove the RAM module(s), clean the contacts carefully, and firmly reinstall them until the retaining clips lock.
2. **Test the RAM individually:** If there are multiple RAM sticks, boot with one stick at a time and try different DIMM slots. If possible, test with a known-good compatible RAM module

3. Task -3

BIOS Setup Utility Practical

Steps:

1. Shut down or restart the PC.
2. Turn it on and repeatedly press **Delete (Del)** or **F2** during startup.

- Common keys: **Del, F2, F10, or Esc.**
3. The **BIOS/UEFI Setup Utility** will open.
 4. On the **Main/System Information** screen, look for:
 - **Motherboard/System Manufacturer**
 - **BIOS Version**

Example observation table:

Item	Example
Motherboard Manufacturer	ASUS
BIOS Manufacturer	American Megatrends (AMI)
BIOS Version	3205
BIOS Mode	UEFI

Result:

The BIOS setup utility was successfully accessed, and the motherboard manufacturer and BIOS version were identified from the main BIOS information screen.

Task-4

CMOS Checksum Error – Defaults Loaded

Meaning:

The message “**CMOS checksum error – Defaults loaded**” means that the BIOS/UEFI detected that the stored CMOS configuration data is incorrect or corrupted. As a result, the system has loaded the **default BIOS settings**. A common cause is a **weak or dead CMOS battery**, which cannot maintain the BIOS settings when the PC is powered off.

How to Resolve It

1. Replace the CMOS battery

- Shut down the PC and disconnect the power.
- Open the computer case.
- Locate the round **CR2032 battery** on the motherboard.
- Remove the old battery and install a new compatible battery.

2. Reset and configure BIOS settings

- Enter the BIOS/UEFI setup using **Del, F2, or the appropriate key** during startup.
- Select **Load Optimized/Setup Defaults** if necessary.
- Correct important settings such as **date/time, boot order, SATA mode, and memory settings**.
- Save the changes and restart the PC.

If the error returns: Check the battery connection and motherboard for problems. If a new battery does not solve it, the motherboard or its CMOS/RTC circuitry may need further diagnosis.

Result: Replacing a failed CMOS battery and restoring the correct BIOS settings should normally resolve the **CMOS checksum error**.